

**Studies on variation in *Rhizoctonia*
bataticola causing dry root rot of chickpea
(*Cicer arietinum* L.)**

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THESIS



Submitted to the

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In

**AGRICULTURE
(PLANT PATHOLOGY)**

By

VISHAL GURAV

Department of Plant Pathology,
Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior
College of Agriculture
Gwalior (M.P.)

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000157

CERTIFICATE - I

This is to certify that the thesis entitled, "**Studies on variation in *Rhizoctonia bataticola* causing dry root rot of chickpea (*Cicer arietinum* L.)**" submitted in partial fulfillment of the requirement for the degree of **MASTER OF SCIENCE in AGRICULTURE (PLANT PATHOLOGY)** of Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior is a record of the bonafied research work carried out by **Mr. Vishal Gurav** under my guidance and supervision. The subject of the thesis has been approved by the Student's Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any degree or diploma (Certificate awarded etc.) or has been published / published part has been fully acknowledged. All the assistance and help received during the course of the investigation has been acknowledged by him.

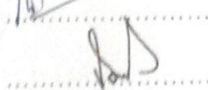

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Member	(Smt. R. Sasode)	
Member	(Dr. S.S. Tomar)	
Member	(Dr. V.B. Singh)	

CERTIFICATE - II

This is to certify that the thesis entitled "**Studies on variation in *Rhizoctonia bataticola* causing dry root rot of chickpea (*Cicer arietinum* L.)**" submitted by **Mr. Vishal Gurav** to the **R.V.S. Krishi Vishwa Vidyalaya, Gwalior** in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE in AGRICULTURE (PLANT PATHOLOGY)** in **Department of Plant Pathology, College of Agriculture, Gwalior** has been, after evaluation, approved by the External Examiner and by the Student's Advisory Committee after an oral examination on the same.

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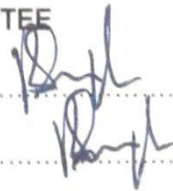
Advisory Committee

MEMBERS OF THE ADVISORY COMMITTEE

Chairman, Dr. (Smt.) Reeti Singh

Head of the Department, Dr. (Smt.) Reeti Singh

Director of Instruction (Dr. H.S. Yadava)



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(Vishal Gurav)

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CHAPTER – 1

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is a premier Rabi season pulse crop of the Indian subcontinent. It is grown over 40 countries across five continents. However, 95% of its area is in developing countries. India grows chickpea on about 8.25 million ha area producing 7.05 million tones which represents 36% and 48% of the national pulse acreage and production, respectively.

Chickpea production has gone up from 3.65 to 7.05 million tones between 1950-51 and 2008-09. During the period, productivity has also increased to 855 kg/ha from 482 kg/ha. Notwithstanding its distribution throughout the country, six states viz., Madhya Pradesh, Rajasthan, Maharashtra, Utter Pradesh, Karnataka and Andhra Pradesh together contribute 91% of the production and 90% of the area of the country. In Madhya Pradesh it occupied an area of 4.76 million ha with a production of 4.78 million tones of grain, the productivity being 1006 kg /ha. (Anon., 2009).

Chickpea contains 17-22% protein, 60-64% carbohydrate and 3-4% fat and is a good source of calcium, phosphorus, vitamins and minerals. The growing leaves of chickpea contain malic acid, which bear medicinal value in intestinal disorders.

In spite of best efforts by the researchers and extension workers, the area under chickpea is gradually declining. Its potential yield is estimated at 5.0 tons/ha, but its global average yield is around 0.8 tons/ha. This huge gap between the potential yield and average of the crop is mainly because the crop is generally grown under rainfed condition and it is very prone toward number of biotic and abiotic stresses prevalent in different parts of the country. Diseases are the most serious constraints to chickpea productivity causing up to 100% losses. It can be affected by more than 50 diseases of varied etiology in different part of world (Nene, 1980). However, only a few of them are divesting to the crop on a world wide basis, of which some of the most important are induced by soil-borne fungi including *Fusarium* wilt, black collar and root rot, dry root rot and *Phytophthora* damping off (Nene and Reddy, 1987).

Among the several soil-borne fungal diseases of chickpea, dry root rot caused by *Rhizoctonia bataticola* Taub. (Butler) is the most severe disease particularly in the central and southern zone, where the crop is mostly grown under rainfed condition. Disease generally appears around flowering and podding stage.

Therefore present piece of study is based on following aspects.

1. Survey and collection of dry root rot infected chickpea plants from villages of Gwalior, Bhind and Morena districts.
2. Isolation, identification and pathogenicity of the isolates collected.
3. Pathogen isolates will be studied for the following characters:-
 - a. Colony color
 - b. Growth rate
 - c. Type of mycelial growth
 - d. Morphological characters.
 - e. Pathological variability.

CHAPTER - II

REVIEW OF LITERATURE

2.1 Historical

The first mention of a plant disease, which may be referred with, certainty to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) in which he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established (1815) the genus *Rhizoctonia*.

In 1924, Small described a *Rhizoctonia*, which developed black sclerotia of variable size and shape in the diseased roots of many woody plants from Uganda and he named it *Rhizoctonia lamrllifera*. Butler (1925), how ever, considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a sclerotium causing root rot of cotton, french beans and cowpeas.

2.2 Taxonomy

Tulasne brothers (1862) studied the taxonomic status of two genus *Rhizoctonia* in 19th century. They published the first extensive morphological study on the genus *Rhizoctonia*. They reduced *Rhizoctonia crocorum* DC and *Rhizoctonia medicaginis* DC to a single species to which they applied a new name *Rhizoctonia vivolaceae* Tul. This reduction to a single form was made after a most careful morphological study of the fungus at all stages.

Among the taxonomic and pathological studies on *Rhizoctonia* in India, the most important is that of Shaw. Before 1912, he had studied two species of *Rhizoctonia*, one of which was characterized by its formulation of small black sclerotia. Alexopoulos and Mims 1983 described the systemic position of *R. bataticola* as follows:-

Kingdom	-	Mycetae.
Division	-	Amastigomycota.
Sub-division	-	Deuteromycotina.
Class	-	Deuteromycetes.
Sub-class	-	Coelomonycetidae.
Order	-	Sphaeropsidales.
Family	-	Sphaeropsidaceae.
Genus	-	<i>Macrophomina</i> .
Species	-	<i>phaseolina</i> .

2.3 Pathogen

The species description of *R. bataticola* was also reported by Subramanian (1971). The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture. On the infected plants sclerotia seen in the pith region and outer surface of roots just below the bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape: globose, oval, oblong, elliptical, curve or even forked and varying in size, but of uniform texture, more or less loosely packed.

The cell of hyphae are barrel shaped, branching more or less at right angles and light to dark brown in colour, composed of free thick-walled cells. The species *Rhizoctonia* is distinguished by the light brown-to-brown hyphae with somewhat loose/spongy sclerotia in the form of bulbils. According to Mehrotra and Aneja (1990) both the species *R. bataticola* and *R. solani* have the variable capacity to produce enzymes.

Jharia (1970) also found that *R. bataticola* is associated with seeds of urid both internally and externally. Thus it seems that *R. bataticola* acts as both soil and seed borne pathogen from Madhya Pradesh.

2.4 Symptoms

Mandal, *et al.* (2002) studied the impact of Hoogly wilt (complex of *R. bataticola* [*Macrophomina phaseolina*], *Fusarium solani* and *Pseudomonas*

solanacearum) of jute, from West Bengal, India in 1949 to 2002. Hoogly symptoms as well as those of root rot and root knot were described by him.

According to Mehrotra and Aneja (1990) the first symptom of the disease was yellowing of the leaves within a day or two and such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants were usually straw colored. The taproot was dark and quite brittle in dry soil.

Singh *et al.* (1990) observed the roots inoculated with *R. bataticola* [*Macrophomina phaseolina*] and reported disintegration of cortical tissues while mycelium and sclerotial bodies plugging the xylem were also seen in longitudinal sections. The sudden collapse of infected plants at the pod-filling stage is might be due to this plugging of the xylem vessels.

2.5 Survey and Surveillance

Chick pea fields in Shimoga, Raichur and Banglore districts in Karnataka, India, were surveyed for the presence of soil-borne pathogens. *Fusarium oxysporum f. sp. ciceri* was detected in all districts. *R. bataticola* (*Macrophomina phaseolina*), which caused wilting and dry root rot disease, was also detected. In Gulbarga and Tengri, *R. bataticola* was the predominant pathogen, infecting 60.0% of the plants in the field (Gurha and Trivedi, 2004).

Shrivastava, *et al.* (2002) conducted a survey in Vindhyan plateau zone of Madhya Pradesh, India during 1998-99 and evaluated the prevalence of wilt and dry root diseases in chickpea. The incidence of *R. bataticola* caused withering and drying of plants that ranged from 8 to 20 per cent.

Reddy *et al.* (1991) stated that the countries like Mandalay, Sayaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola*. Anurag *et al.* (1992) observed that dry root rot caused by *R. bataticola* was major disease of chickpea.

2.6 Occurrence and losses

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta

et al. (1983). Root rot was also reported as the most common diseases in south East Asian countries and was main constraint in chickpea production (Anon. 1986, FAO, 1990).

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot (*R. bataticola*) in Southern Haryana and reported that the incidence was high from late October to mid November which decreased in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than at 7.5 cm.

The *R. bataticola* was reported to be responsible for serious diseases like stem blight, foliage blight, root rot, seedling blight, charcoal rot, wilt, collar rot, dry rot, pod and seed rot (Raut and Ingle, 1989) in several crops such as soybean, gram mung, urid, fruit trees and many vegetables (Mukharji and Bhasin, 1986). This sclerotial fungus is of wide geographical distribution ranging from Formosa, Philippines, India, Ceylon, East Africa, Palestine and Egypt to the West Indies and the Eastern United State (Ashby, 1927). Significant reduction (25-70%) in chickpea production was caused by this disease in India (Ahmed et al. 1986 and Pandey and Singh, 1990).

2.7 Pathogenicity

An experiment was conducted to differentiate isolates of cotton root rot pathogens (23 isolates of *Rhizoctonia solani* and 25 isolates for *R. bataticola*[*Macrophomina phaseolina*]) collected from north India using RAPD markers and to study the relationship between molecular variability and pathogenicity. Cultural characteristics varied for *R. bataticola* isolates. (Monga, et al., 2007)

Anilkumar and Sastry (1980) reported the variation among eight isolates of *Rhizoctonia bataticola* from different hosts with regard to growth rate, effect of media, sclerotial number and size, effect of inoculation temperature, fungicides and virulence.

Chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth, i.e., early seedling stage and flowering stage. The wilting was due to *F. oxysporum* f. sp. *ciceri*, dry

root by *R. bataticola* and *Sclerotium rolfsii* caused collar rot. (Nema and Khare, 1973)

Dermiro *et al.* (1994) observed that dry root rot disease of gram has also become a serious disease affecting productivity of this crop. They isolated *R. bataticola* from plants and seeds of chickpea cultivars Azizive 94 and proved the pathogenicity.

2.8 Variability

Cultural, Morphological and Pathogenic Variability

Agakhani and Dubey (2009) collected twenty three isolates of *R. bataticola* causing dry root rot of chickpea from 10 different major chickpea growing states of India. They were highly variable in their morphological and cultural characters as well as pathogenicity / virulence. The most virulent isolate (RB1 from Karnataka) was fast growing and produced largest sclerotia.

Kanchan and Biswas (2009) stated that the nature of mycelial was varied from fluffy dark brown to partially fluffy mycelium colony, with smooth margin. The potency of infection of different isolates of *R. bataticola* was varied on four different cultivars of pigeon pea. The early age group of plant showed low disease severity as compared to middle and above middle age group.

Seven isolates of *M. phaseolina* incident of charcoal rot of maize, obtained from different agro-ecological zones of India was studied and found that they varied in their cultural characteristics and pathogenic behavior. On the basis of colony colour, isolates were divided into four groups i.e., grayish white, blackish gray, dark black in center periphery cremish and cottony white color. (Shekhar *et al.*, 2006)

Byadgi and Hegde (1985) studied the variation in isolates of *R. bataticola* obtained from six different plant species differed in their cultural characters, growth rate, morphology of the sclerotia, pycnidia and pycnidospores and virulence. Cultural characters of the isolates after 3 days of incubation indicated bean isolate grows significantly faster followed by Bengal gram and cowpea isolates, which produced fluffy dark brown mycelium. The sorghum and soybean isolates did not differed significantly and exhibited moderate growth with the partially fluffy mycelium. The *Gliricidia* isolate produced poor growth and

appressed light brown mycelium. The colonies had smooth margins in all the cases.

Jharia (1985) isolated *R. bataticola* from three different sources. He observed variation in their growth rate. The soil isolate was fast growing (90mm) followed by root (89mm) and seed (67 mm) isolates on potato dextrose agar medium. Sobti and Bansal (1988) studied variation in cultural characteristics of root, leaf and seed isolates of *R. bataticola in-vitro* tests on different media, the seed and root isolates had the smallest and large sclerotia, respectively. Growth characteristics of potato dextrose agar, Richard's medium and groundnut kernel agar is also described.

Mandal *et al.* (1998) studied the sclerotial morphology and pathological reactions of 13 isolates of *M. phaseolina* from jute growing regions of India. He found that sclerotia size ranged from 66.14 to 128.25 μm and the color varied from light to dark brown. Sclerotial shape in most cases was irregular except for few which were round to elongate.

Ratnoo *et al.* (1997) studied two isolates of *M. phaseolina* from cowpea roots and stem showed variability in their cultural characteristics, M1 isolate showed dark grey mycelium, aerial fluffy growth, hyphae thin become thicker with age, septate and profusely branched. The sclerotial formation was also abundant, blackish, globose to round and measured 49-76 μm in diameter whereas in M-2 isolate, it exhibited grey, aerial cottony, compact regular and fast growth of mycelium, hyphae thin, septate branching to form bunch like structure.

Sobti and Sharma (1992) studied 13 isolates of *R. bataticola* of groundnut collected from Rajasthan. Cultural, morphological characteristics and virulence of the isolates were investigated and reported that hyphal cell size ranged from 15.30-24.64 X 4.16-8.48 μm and sclerotial size ranged from 73.88-188.33 X 51.09-155.90 μm and the colony diameter of the isolates ranged from 59.00 to 84.00 mm.

Fourteen isolates of *M. phaseolina* obtained from different plant species and plant parts of the same host differed in their culture characteristics and pathogenic behavior (Hooda and Grover, 1982). The isolates exhibited wide variation in mycelial dry weight as well as sclerotial production. Among the

thirteen isolates from four leguminous crops, least growth was noted in isolates from mung bean stem, mung bean pods and cowpea leaf, whereas mung bean leaf and seed isolates showed significantly higher growth than others. Mung bean seed isolate had very fluffy growth. Castor isolate had significantly higher mycelial dry weight than other isolates, but produced few sclerotia. There was no correlation between the sclerotial production and mycelial dry weight among the isolates obtained from the same host tissues.

Monga and Sheo Raj (1994) observed thirteen isolates of *R. solani* and categorized into four distinct groups on the basis of cultural characteristics. The cultural characteristics of different isolates revealed that isolate in group Rs-2 was faster in mycelial growth which was significantly more than any other isolate belonging to other groups. However, isolates in Rs-1 and Rs-3 did not show significant growth differences among themselves.

Five different isolates of *R. solani* on the basis of pathogenic behavior. All isolates were pathogenic on all the five varieties tested but they differed in their degree of virulence. The isolates A and B were more virulent than the isolates C, D and E. Isolate E was found to be the least virulent among all the isolates tested (Singh *et. al.*, 1990).

According to Raut and Ingle (1989) the isolates of *R. bataticola* from different cultivated crops belonging to seven families varied in their growth rate in terms of size of sclerotia, production of pycnidia and size of pycnidiospores. Differences were evident even between the isolates of crop belonging to same family. Light was found essential for pycnidia formation. Comparatively cotton and cowpea isolates formed more sclerotia. The size of sclerotia varied with the host. The sclerotia of cotton isolate were the largest (mean dia. 115.06 μm) and soybean isolates the smallest (56.33 μm). Cluster bean, pigeon pea and black gram isolates measured between 61 - 70 μm , safflower, okra, green gram, chilli, sunflower and citrus isolates between 71 - 80 μm and cow pea, sorghum, chickpea and sesamum isolates between 81 - 90 μm .

The variation among eight isolates of *R. bataticola* from different hosts with regards to growth rate, effect of media, sclerotial number and size, effect of

inoculation temperature, fungicides and virulence was reported by Anilkumar and Sastry (1980).

Jain *et al.* (1973) found *R. bataticola* associated with root, stem, leaf, pod and seeds of *Phaseolus mungo*. The isolates from various plant parts and soil showed differences in virulence. The soil isolate was most pathogenic. The isolates differed in their growth pattern and sclerotial size. The leaf isolate developed the largest sclerotia and the seed and soil isolates the smallest. When grown on different media the isolates varied in growth pattern and growth rate.

Dhingra and Sinclair (1972) studied 15 isolates of *M. phaseolina* and reported morphological variations in isolates from root, pod and seed of soybean plants. They observed that sclerotia from root isolates ranged from 122 – 188 x 77 – 127 μm , while stem isolates 84 – 100 x 60 – 71 μm .

Critopoulos (1953) compared two isolates of *R. bataticola* and reported variation in their morphological characteristics. He found that in one isolate, sclerotia were spherical to ellipsoid 60-165 X 57-114 μm , while in the others round to oval measuring 36-99 X 36-81 μm .

CHAPTER - III

MATERIALS AND METHODS

Present investigations were undertaken to study the "Variability in isolates of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot in chickpea (*Cicer arietinum* L.)." The materials used and methods followed are described below;

EXPERIMENTAL SITE

The present studies were conducted in the Department of Plant Pathology and Research farm of College of Agriculture, Gwalior during Rabi season of 2009-10. Gwalior is situated in Northern part of Madhya Pradesh at an elevation of 211.52 meters from mean sea level and lies between altitude and longitude 26° 14, North and 78°15, East, respectively.

Climate

The climate of Gwalior is subtropical. The rainy season normally starts from middle of June after commencement of monsoon and last up to September. Maximum precipitation of rains occurred in month of July and August. Summer season from April to mid June. October is the transitory month between rain and winter season.

Soil

The soil of experimental site was alluvial clay loam texture and pH of 7.6. The available nitrogen, phosphorus and potassium were low, medium low to medium, respectively. The water holding capacity of soils is also medium.

A. MATERIAL

1. General

Borosil make glassware's were used throughout the investigation. These were cleaned by through washings with detergent powder and subsequent rinsing with water. Glassware's were sterilized in hot air oven at 180⁰ C for two hours.

The inoculation needle and other metallic instruments were sterilized by dipping them in an alcohol and heating over flame. The culture media were sterilized in an autoclave at 1.05 kg pressure per square inch. (15 lbs, p.s.i / at 121.6⁰C) for 15 minutes. At the time of pouring of plates, streptomycin sulphate @ 600 mg/l was added to the medium to inhibit the bacterial growth. The soil/ sand and chickpea straw were sterilized at 2.1 kg pressure per square inch (30 lbs pressure p.s.i) for two hours.

2. Test pathogen

Rhizoctonia bataticola the causal organism of chickpea dry root rot.

3. Culture Media

The following media were used during the course of investigation.

(a) Potato dextrose agar (P.D.A)

Peeled potato	-	200 g
Dextrose	-	20.0 g
Agar-agar	-	20.0 g
Distilled water	-	1000 ml

(b) Potato dextrose broth (P.D.B.)

Peeled potato	-	200 g
Dextrose	-	20.0 g
Distilled water	-	1000 ml

For the preparation of PDA, potato slice was boiled in 500 ml water until they become soft. The supernatant was filtered. The supernatant and melted agar was then mixed, measured and the volume was restored to 1000 ml with hot water. It was again boiled for 5 minutes and then dextrose was added. The medium was

poured into flask and culture tubes. The mouth of flask and culture tubes was plugged with cotton plug and sterilized in an autoclave at the pressure of 15 pound/square inch (1.05 kg/cm²) for 15 minutes and temperature goes up to 121.6°C which is sufficient for proper sterilization. After this the medium was used under aseptic condition for bioassay of the test fungus.

In PD broth similar procedure as described above was employed only agar agar was not added in potato dextrose broth.

B. METHODS

1. Survey of root rot of chickpea

Chickpea fields of Gwalior, Bhind and Morena districts were surveyed to find out the incidence of root rot during Rabi season of 2010. Five locations from Gwalior (Research Farm, Badagaon, Utila, Berja, Rayru), five from Bhind (Rampura, Dandraua, Mehgaon, Piproli, Gormi) and Morena (Ambah, Dimni, Khera, Morenagaon, Shanichara) were randomly selected. Thus total of 15 localities were selected and in each locality 5 fields were randomly selected. In each field the root rot incidence was randomly recorded on randomly selected 1m² patches. The root rot incidence on the plants selected patches was recorded with the help of following formula:-

$$\text{Root rot incidence (\%)} = \frac{\text{Root rot infected plants}}{\text{Total number of plants}} \times 100$$

The localities wise incidence of root rot was worked out by calculating the mean of five fields of respective locality. District wise incidence of root rot was calculating the average of five localities of respective district (Table 1).

Table 1(a): Collection of diseased plant / root samples of chickpea, collected from different districts of Northern Madhya Pradesh

State	Isolates no.	Isolate Code	Districts	Village/Locations
Madhya Pradesh	1	GWL ₁	Gwalior	Research Farm
	2	GWL ₂	Gwalior	Badagaon
	3	GWL ₃	Gwalior	Utila
	4	GWL ₄	Gwalior	Berja
	5	GWL ₅	Gwalior	Rayru
	6	BND ₁	Bhind	Rampura
	7	BND ₂	Bhind	Dandraua
	8	BND ₃	Bhind	Mehgaon
	9	BND ₄	Bhind	Piproli
	10	BND ₅	Bhind	Gormi
	11	MRN ₁	Morena	Ambah
	12	MRN ₂	Morena	Dimni
	13	MRN ₃	Morena	Khera
	14	MRN ₄	Morena	Morenagaon
	15	MRN ₅	Morena	Shanichara

Table 1 (b)

S.No.	State	Districts	Number of locations
1	Madhya Pradesh.	Gwalior	5
2		Bhind	5
3		Morena	5
Total			15

2. Isolation of *Rhizoctonia bataticola* from plant samples

The roots of diseased chickpea plant showing root rot symptoms were washed thoroughly with tap water, small pieces from infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl_2) solution for one minute followed by three changes in sterilized distal water to remove trace of HgCl_2 . The pieces were then transferred aseptically to petri plates containing PDA. Inoculated petri plates were incubated at $28 \pm 1^\circ\text{C}$ for three to five days and examined at frequent intervals to see the growth of the fungus developing from different pieces.

3. Purification and identification of isolates of *Rhizoctonia bataticola*

The cultures of *R. bataticola* were purified by sub-culturing the single hyphal tip method and maintained by mass transfer on potato dextrose agar medium at room temperature. After purification these isolates of *R. bataticola* were identified by observing the colony against light with the naked eyes and later confirmed with the help of microscope.

4. Variability in isolates of *Rhizoctonia bataticola*

4.1 Cultural variability

The cultural characters of isolates of *R. bataticola* were recorded from culture grown on PDA. Twenty ml of PDA was poured in each of previously sterilized petri plates. Five mm discs were cut through sterilized cork borer from the margin of seven days old colony of the fungal culture grown in petri plates. One disc was placed in the centre of each plate and incubated at $28 \pm 1^\circ\text{C}$ for seven days. Three replications were maintained for each isolate. The differences between observations

regarding colony color, Growth rate and type of mycelial growth of each isolate were taken.

4.2 Morphological variability:

The slides of 15 isolates were prepared in lactophenol from 10 days old culture. For morphological studies, 10 observations for size of sclerotia and size of hyphal width and number of sclerotia per microscopic field belonging to each isolates were taken under high power (40x) microscopic field. The differences in color of sclerotia and shape of sclerotia were observed. These isolates were categorized in various groups according to size of hyphal width, size of sclerotia, number of sclerotia, colour of sclerotia and shape of sclerotia.

4.3 Pathological variability:

4.3.1 Pathogenicity test

Blotter paper technique as described by Nene *et al.* (1981): -

To conduct the pathogenicity test, 20 surface sterilized (5 min in 2.5% sodium hypo chloride) seeds of the test line of susceptible variety JG 362 were sown in autoclaved riverbed sand in pots. The two-mycelial mats of *R. bataticola* were grown on P.D. Broth media for 5 days the then removed in 100 ml. sterilized distilled water, macerated in a warring blender for 1 min. The five days old seedlings were uprooted and washed the root system in running water followed by rinsing in sterile distilled water, and then dipped in inoculum solution for 30 seconds up and down. After that, inoculated seedlings were placed side by side on a blotter paper in two replications on tray and also maintain the line of control without culture. Placed the tray in an incubator at 35° C for 8 days and provided 12 hours artificial light. Moistened the blotters adequately every day or alternate day. At the end of incubation period (8 days), examined the seedlings for the disease or damage development.

5 Statistical Analysis:

Design

The Statistical model design in single environment was used as defined by Chandel (1999).

$$Y_{ijk} = k + t_i + b_{ij} + e_{ijk}$$

Where,

k = the general

t_i = the effect of i^{th} treatment

b_{ij} = the effect of j^{th} replication in i^{th} treatment

e_{ijk} = the error associated with each observation.

The skeleton of analysis of variance for randomized complete block design is given below:

Source of Variance	d.f.	Observation mean sum of squares	Expected mean sum of square
Replication	$r - 1$	$M.S_1$	$2_{ei} + 2_{tri}$
Treatment	$t - 1$	$M.S_2$	$2_{ei} + 2_{rti}$
Error	$(r - 1)(t - 1)$	$M.S_3$	2_{ei}
Total	$rt - 1$		

CHAPTER - 4

EXPERIMENTAL FINDINGS

1. Survey of Root Rot of Chickpea:

For knowing the prevalence of the disease on farmer's field a planned survey of the Chickpea growing area was carried out in 2010. Cultivator's fields were surveyed for recording the incidence of root rot in villages of Gwalior, Bhind and Morena districts at flowering and podding stage. In all the locations surveyed none of the field remained free from root rot disease.

It is evident from Table 2 that incidence of root rot in the surveyed fields ranged from 5.66 to 22.77 per cent.

The minimum incidence of root rot was observed in Rayru (5.66%) followed by Khera (6.44%), Badagaon (7.24%), Berja (8.22%), Ambah (8.72%) and Gormi (9.76%).

The maximum incidence of root rot was observed in Dandraua (22.77%), followed by Utila (18.90%), Mehgaon (18.22%), Piproli (16.69%), Dimni (15.70%), Research farm (15.30%), Rampura (15.16%), Morenagaon (14.50%) and Shanichara (13.05%).

Chickpea plants showing symptoms of dry root rot were collected during the survey (Fig-1).

Table 2: Occurrence of Root Rot of chickpea in different locations Gird zone of Madhya Pradesh

S. No.	Location	Season	Soil Type	Stage of Crop	Disease incidence (%)*
Gwalior					
1	Research Farm	Rabi	Sandy Loam	Flowering	15.30
2	Badagaon	Rabi	Sandy Loam	Podding	7.24
3	Utila	Rabi	Black	Flowering	18.90
4	Berja	Rabi	Sandy Loam	Podding	8.22
5	Rayru	Rabi	Black	Podding	5.66
Bhind					
6	Rampura	Rabi	Sandy Loam	Podding	15.16
7	Dandraua	Rabi	Sandy Loam	Flowering	22.77
8	Mehgaon	Rabi	Black	Flowering	18.22
9	Piproli	Rabi	Sandy Loam	Podding	16.69
10	Gormi	Rabi	Black	Flowering	9.76
Morena					
11	Ambah	Rabi	Sandy Loam	Podding	8.72
12	Dimni	Rabi	Black	Podding	15.70
13	Khera	Rabi	Sandy Loam	Flowering	6.44
14	Morenagaon	Rabi	Sandy Loam	Podding	14.50
15	Shanichara	Rabi	Black	Flowering	13.05

* Average of 5 locations.

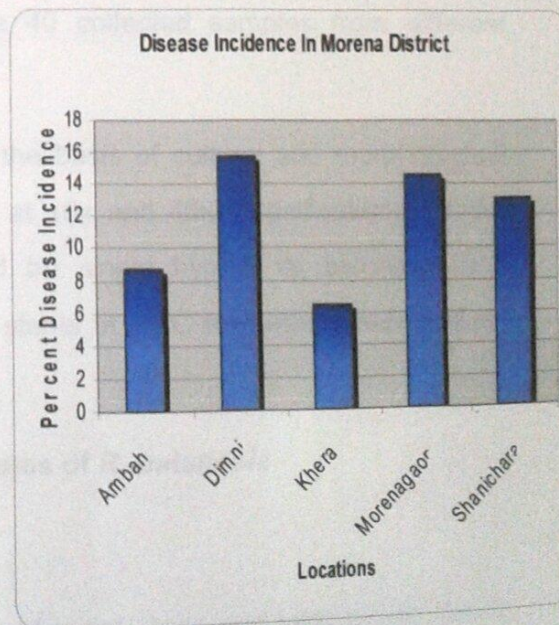
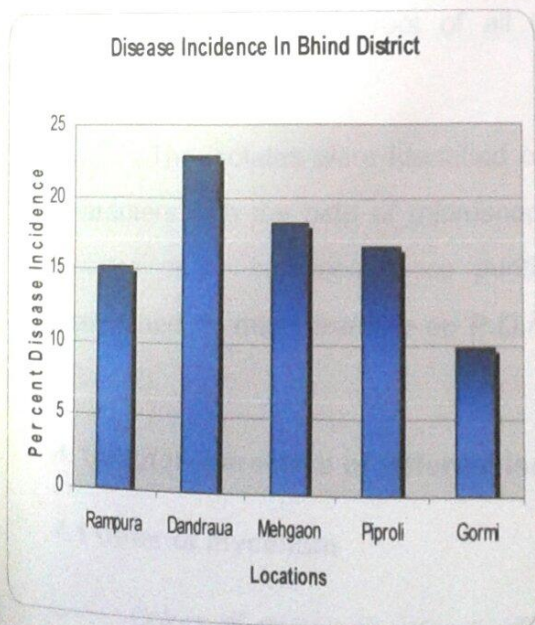
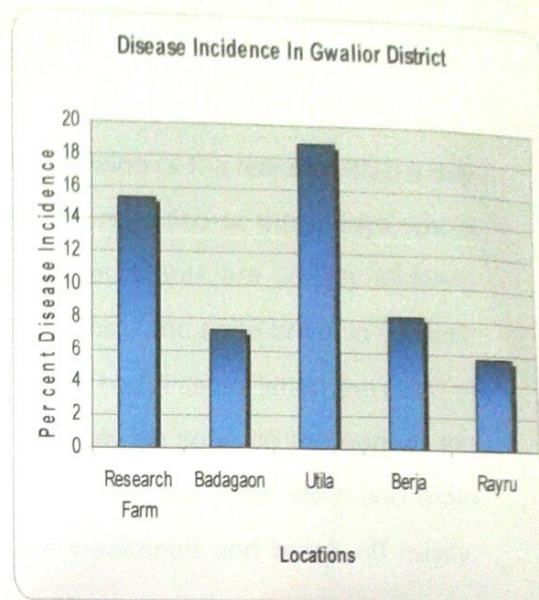
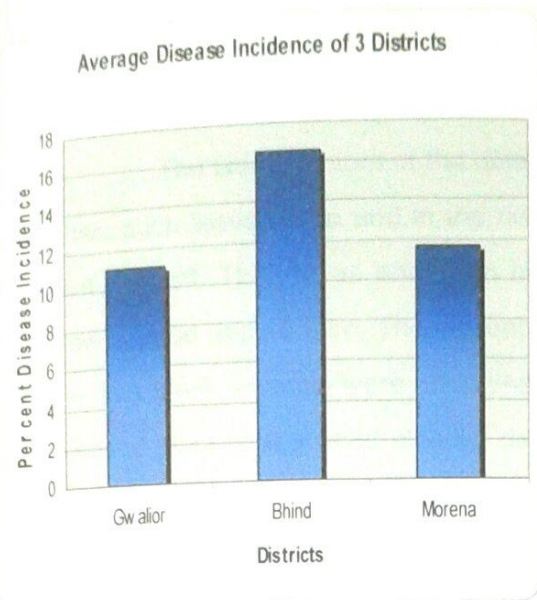


Fig. 1: Survey of Dry Root Rot of Chickpea.

2. Symptomatology

The first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually showing straw colored appearance. The taproot becomes dark and quite brittle in dry soil. The dark black sclerotia were observed and within host tissues when the stem of diseased plant is examined closely, dark lesion may be seen on the bark at the ground level. On uprooting of the plants from the soil, the basal stem and main root shows dry rot symptoms. The tissues are weakened and break off easily. (Plate-1)

3. Isolation, purification and identification of *Rhizoctonia bataticola*

Isolation from infected chickpea plants showing symptoms of “Dry root rot” were made on potato dextrose agar medium. *R. bataticola* was found to be associated with plant / root of all the 40 collected samples from different localities.

The isolates were identified on the basis of cultural and morphological characters with the help of microscope at 10x and 40x magnifications. All the isolates of *R. bataticola* were purified by single hyphal tip technique and maintained by mass transfer on P.D.A. slants at 25°C for further investigations (Plate-2).

4. Cultural characters of different isolates of *R. bataticola*

4.1 Color of mycelium

Colors of mycelium were hyaline at initial stage and later turned brown, dark brown or black. Details of color of mycelium for all 15 isolates after 7th day of incubation are given in Table 3 (Plate-3)



Plate 1: Chickpea plant showing symptoms of Dry Root Rot.



On Potato Dextrose Agar



On Potato Dextrose Broth

Plate 2: Growth of *Rhizoctonia bataticola*

Table 3: Color of mycelial growth of various isolates of *Rhizoctonia bataticola*

Isolates No.	Color of Mycelium
1	Dark brown with white margin
2	Black
3	Black with crinkling at centre
4	Dark brown
5	Brown with white margin
6	Black with white fluffy growth at centre
7	Dark brown
8	Brown with crinkling at centre
9	Dark brown with white fluffy growth at centre
10	Black
11	Dark brown with crinkling at centre
12	Black with white margin
13	Dark brown with white fluffy growth at centre
14	Brown with crinkling at centre
15	Dark brown

Table 4: Grouping of isolates of *R. bataticola* on the basis of mycelium color

Characters	Isolates	Total	Per cent
Brown	5,8,14	3	20
Dark brown	1,4,7,9,11,13,15	7	46.67
Black	2,3,6,10,12	5	33.33
	Total	15	100

Color of mycelium was observed and categorized into three groups as brown, dark brown and black. Out of the 15 isolates 3 (20%) Isolates were found as brown, 7 (46.67%) isolates were found as dark brown and 5 (33.33%) isolates were found as black color of mycelium.

4.2 Type of mycelial growth on P.D.A.

Cultural variability were studied on P.D.A medium and the characters observed on type of mycelial growth and growth rate of isolates of *Rhizoctonia bataticola* were recorded on alternate days after 6 days of incubation at $25\pm 1^{\circ}\text{C}$ (Table 5, 6, 7 and Plate-4).

On the basis of type of mycelial growth the isolates were categorized as fluffy, partial submerged and submerged. It is evident from the data presented in Table 3 that out of 15 isolates, 3 (20%) isolates were fluffy, 7 (46.67%) isolates were partial submerged and 5 (33.33%) isolates were having submerged type of mycelial growth (Table 6).

Table 5: Type of mycelial growth and growth rate of various isolates of *Rhizoctonia bataticola* on potato dextrose agar medium at different intervals

Isolates No.	Type of mycelial growth	Growth rate (mm / day) *					Average growth per day
		1	2	3	4	5	
1	Submerged	9	23	38	55	71	14.2
2	Fluffy	0	7	18	34	56	11.2
3	Fluffy	0	5	15	30	49	9.8
4	Submerged	5	13	27	42	61	12.2
5	Partial submerged	13	28	44	57	74	14.8
6	Partial submerged	18	31	47	62	78	15.6
7	Submerged	23	37	57	72	90	18.0
8	Partial submerged	11	25	42	57	74	14.8
9	Partial submerged	13	29	47	60	77	17.4
10	Partial submerged	21	34	50	65	83	16.6
11	Submerged	0	9	22	40	59	11.8
12	Partial submerged	25	42	61	78	90	18.0
13	Partial submerged	12	23	38	52	71	14.2
14	Fluffy	7	18	31	46	64	12.8
15	Submerged	0	8	20	37	58	11.6

* Average of three replications

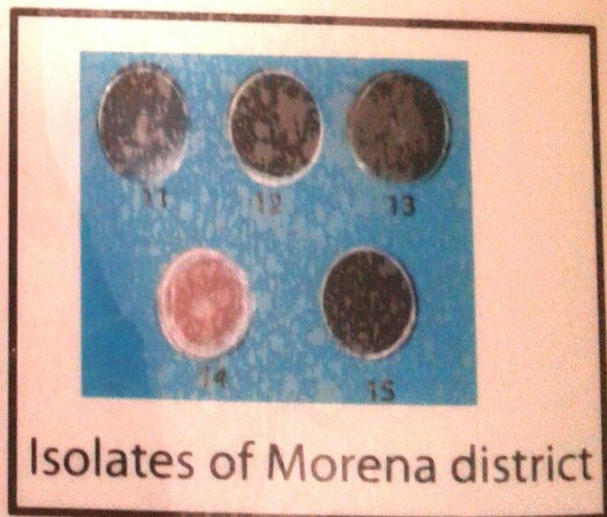
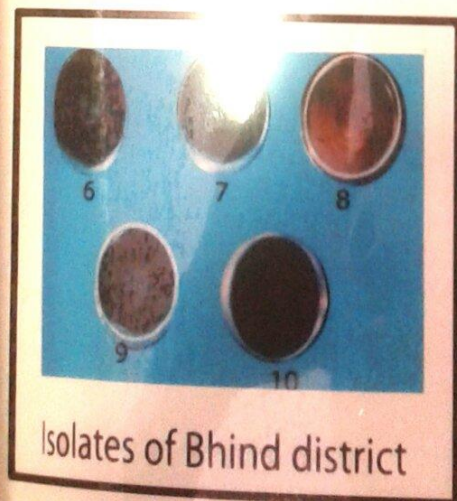
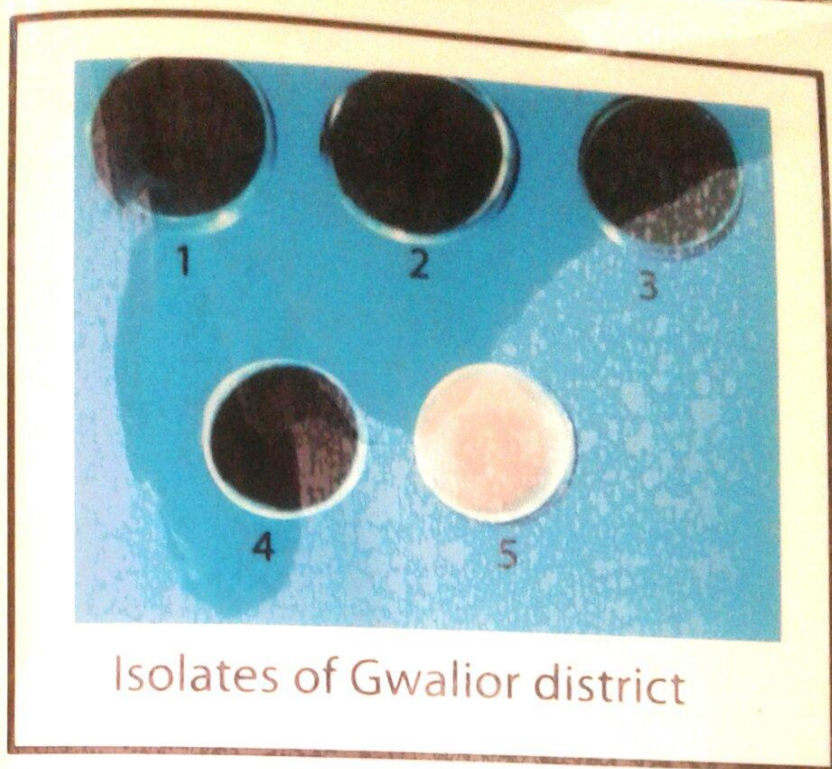


Plate 3: Color variation in various isolates of *Rhizoctonia bataticola*

Table 6: Grouping of isolates of *R. bataticola* on the basis of type of mycelial growth

Mycelial growth	Isolates	Total	Percent
Fluffy	2, 3, 14	3	2
Partial submerged	5, 6, 8, 9, 10, 12, 13	7	46.67
Submerged	1, 4, 7, 11, 15	5	33.33
Total		15	100

4.2 Growth rate on P.D.A.

Studies on growth rate of isolates were conducted to know the fast growing, medium growing and slow growing isolates at different intervals on P.D.A. Observations on growth rate of different isolates were made after each day up to 5th day of incubation. On the basis of average growth per day the data denotes that the slow growth (9.5-12.5 mm) has been obtained in isolates 2, 3, 11, 15 and isolates 1, 4, 5, 8, 13, 14 showed medium growth (12.6-15.5 mm). Rest of the isolates attained the fast growth (15.6-18.0 mm). On the basis of 2nd day observation among all the isolates, the isolate no. 12 belong to Dimni had maximum growth 42 mm. while the minimum growth 5 mm. was observed in isolate no. 3 which is from Utila (Table 5).

Table 7: Grouping of isolates of *R. bataticola* on the basis of growth rate

Characters	Isolates	Total	Per cent
Slow growth (9.5-12.5 mm)	2, 3, 11, 15	4	26.7
Medium growth (12.6-15.5 mm)	1, 4, 5, 8, 13, 14	6	40
Fast growth (15.6-18.0 mm)	6, 7, 9, 10, 12	5	33.33
	Total	15	100



Fluffy Growth : Isolate No. 2



Partial Submerged : Isolate No. 8



Submerged Growth : Isolate No. 4

Plate - 4 Type of mycelial growth of various isolates of *Rhizoctonia bataticola* on potato dextrose agar medium.

5. Morphological characters of different isolates of *Rhizoctonia bataticola*

To know the variability in morphological characters of different isolates of *Rhizoctonia bataticola* viz., size of hyphal width, size of sclerotia, number of sclerotia colour of sclerotia and shape of sclerotia have been studied and summarized in Table. – 8. Ten Microscopic observations under high power (40 X) were carried out for each isolate and the average value is presented (Table 8, 9, 10,11,12,13 and Plate-3).

Table 8: Morphological characters of various isolate of *Rhizoctonia bataticola**

Isolate No.	Size of hyphal width (µm)	Size of Sclerotia (µm)**	Number of Sclerotia / microscopic field	Colour of Sclerotia	Shape of Sclerotia
1	3.5	28.7	89.6 x 37.0	Dark brown	Oval
2	4.2	6.7	50.0 x 44.0	Black	Round
3	4.0	19.5	86.3 x 57.6	Black	Oval
4	3.7	8.5	69.0 x 52.3	Dark brown	Oval
5	3.7	19.2	27.0 x 24.0	Brown	Round
6	4.2	16.5	33.0 x 31.0	Black	Round
7	3.5	19.0	46.3 x 43.6	Dark brown	Round
8	5.0	55.0	37.3 x 30.0	Brown	Round
9	3.0	28.7	49.3 x 43.0	Dark brown	Round
10	3.2	20.2	36.6 x 31.3	Black	Round
11	5.7	14.5	44.6 x 41.3	Dark brown	Round
12	4.5	15.2	90.0 x 81.6	Black	Round
13	4.0	25.7	41.0 x 26.3	Dark brown	Oval
14	4.5	29.0	49.0 x 41.3	Brown	Round
15	4.7	13.2	54.0 x 49.6	Dark brown	Round

* Average of 10 observations.

** Observed under low power (10 X)

5.1 Size of hyphal width

It is evident from the data presented in Table 8 that wide range of variations has been observed in size of hyphal width.

Average of 10 microscopic observations on the size of hyphal width varied from 3-5.7 μ m. On the basis of observations on size of hyphal width the isolates were classified into three categories viz., very thin (3.0-3.8 μ m), thin (3.9-4.7 μ m) and thick (4.8-5.7 μ m). Out of 15 isolates, 6 (40%) isolates come under very thin size of hyphal width, 7 (46.67%) isolates comes under thin size of hyphal growth, and 2 (13.33%) isolates comes under thick size of hyphal growth. Among all the isolates, the isolate no. 11 belong to Ambah had maximum size of hyphal width 5.7 μ m. while the minimum size of hyphal width 3.0 μ m. was observed in isolate no. 9 which is belong to Piproli. The *Rhizoctonia* isolates consist of barrel shaped cells, mycelium branched and light to dark brown in color, septation within the cell also varied from one isolate to another isolate. (Table 9) Plate-5.

Table 9: Grouping of isolates of *R. bataticola* on the basis of size of hyphal width

Characters	Isolates	Total	Per cent
Very thin (3-3.8 μ m)	1, 4, 5, 7, 9, 10	6	40
Thin (3.9-4.7 μ m)	2, 3, 6, 12, 13, 14, 15	7	46.67
Thick (4.7-5.7 μ m)	8, 11	2	13.33
	Total	15	100

5.2 Size of sclerotia

The data presented in Table 8 denote that on the basis of 10 Microscopic observations the size of sclerotia varied from 27 x 24 – 90.0 x 81.6 μ m. The isolates were classified into three groups, viz; minute (27.0-47.0 x 24.0-43.2 μ m), medium size (47.1-68.0 x 43.3-62.4 μ m) and bold size (68.1-90.0 x 63.5-81.6 μ m). Out of 15 isolates, 7 (46.66%) of isolates were minute, 4 (26.67%) isolates were medium size and 4 (26.67%) were of bold size sclerotia. Among all the isolates, the isolate no. 12

belong to Dimni had maximum bold size of sclerotia 90.0 X 81.6 μm . while the very Minute size of sclerotia 24 X 81.6 μm . was observed in isolate no. 5 which is belong to Rayru (Table 10) Plate-6.

Table 10: Grouping of isolates of *Rhizoctonia bataticola* on the basis of size of sclerotia

Characters	Isolates	Total	Per cent
Minute (27.0-47.0 x 24.0-43.2 μm)	5, 6, 7, 8, 10, 11, 13	7	46.66
Medium (47.1-68.0 x 43.3-62.4 μm)	2, 9, 14, 15	4	26.67
Bold (68.1-90.0 x 63.5-81.6 μm)	1, 3, 4, 12	4	26.67
	Total	15	100

5.3 Number of sclerotia

Number of sclerotia per microscopic field was also observed for each isolate and grouped into, three groups viz; minimum (5-21), medium (22-38), maximum (39-55). The number of sclerotia varied from 6.7 to 55.0. Out of 15 isolates 10 (66.67%) isolates were observed under minimum number of sclerotia, 4 (26.67%) isolates were encountered under medium number of sclerotia and 1 (6.66%) isolates were categorized under maximum number of sclerotia. Among all the isolates, the isolate no. 8 belong to Mehgaon had maximum number of sclerotia 55.0 while the minimum number of sclerotia 6.7 was observed in isolate no. 2 which is belong to Badagaon (Table 11, Plate-7).

Table 11: Grouping of isolates of *Rhizoctonia bataticola* on the basis of number of sclerotia

Characters	Isolates	Total	Per cent
Minimum (5-21)	2, 3, 4, 5, 6, 7, 10, 11, 12, 15	10	66.67
Medium (22-38)	1, 9, 13, 14	4	26.67

Maximum (39-55)	8	1	26.67
	Total	15	100

5.4 Color of sclerotia

Color of sclerotia was also observed and categorized into three groups as brown, dark brown and black. Out of the 15 isolates 3 (20%) Isolates were found as brown, 7 (46.67%) isolates were found as dark brown and 5 (33.33%) isolates were found as black color of sclerotia (Table 12, Plate-8).

Table 12: Grouping of isolates of *Rhizoctonia bataticola* on the basis of color of sclerotia

Characters	Isolates	Total	Per cent
Brown	5,8,14	3	20
Dark brown	1,4,7,9,11,13,15	7	46.67
Black	2,3,6,10,12	5	33.33
	Total	15	100

5.5 Shape of sclerotia

Shape of sclerotia was also observed for each isolates and categorized into two groups as oblong shape and round shape. Out of the 15 isolates 4 (26.67%) Isolates were found as oblong shape and 11 (73.33%) isolates were found as round shape of sclerotia (Table 13, Plate-9).

Table 13: Grouping of isolates of *Rhizoctonia bataticola* on the basis of shape of sclerotia

Characters	Isolates	Total	Per cent
Oblong shape	1, 3, 4, 13	4	26.67
Round shape	2, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15	11	73.33
	Total	15	100



(I) Very thin:- Isolate no. 4.



(II) Thin:- Isolate no. 12.



(III) Thick:- Isolate no. 11.

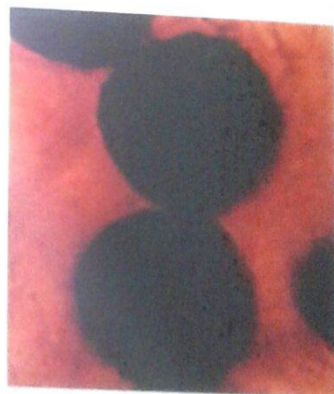
Plate 5: Variation in size of hyphal width



(I) Minute:- isolate no. - 7



(II) Medium:- isolate no.- 2



(III) Bold:- isolate no.- 12

Plate 6: Variation in size of sclerotia



(I) Minimum: - isolate no.- 3

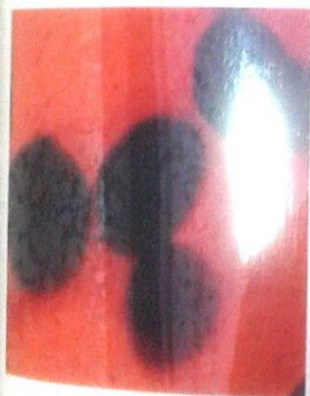


(II) Medium: - isolate no.- 14

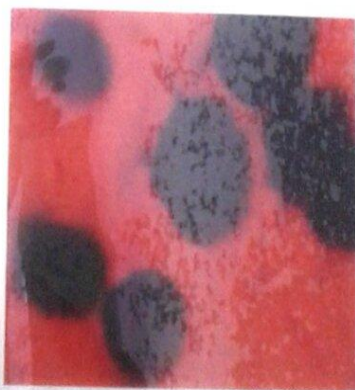


(III) Maximum: - isolate no.- 8

Plate 7: Variation in number of sclerotia



(I) Brown color:- isolate no.- 5



(II) Dark brown:- isolate no.- 9



(III) Black color:- isolate no.- 12

Plate 8: Variation in colour of sclerotia



(I) Round shape: - isolate no.- 8



(II) Oblong shape: - isolate no.- 3

Plate 9: Variation in shape of sclerotia

6. Pathological variability:

6.1 Pathogenicity test

Blotter paper technique as described by Nene *et al.* (1981) was employed to know the pathological variability among the isolates of *Rhizoctonia bataticola*. A highly susceptible chickpea variety JG-365 was used to see the disease rating and maximum mortality causing after inoculation by the isolates (Table 14).

Table 14: Pathological characters of various isolate of *R. bataticola*

Isolates no.	Disease rating	Category
1	8.1	Highly virulent
2	7.3	Virulent
3	6.6	Virulent
4	7.8	Highly virulent
5	7.9	Highly virulent
6	8.7	Highly virulent
7	8.6	Highly virulent
8	7.7	Virulent
9	8.2	Highly virulent
10	7.9	Highly virulent
11	6.7	Virulent
12	8.8	Highly virulent
13	7.3	Virulent
14	7.8	Highly virulent
15	6.9	Virulent

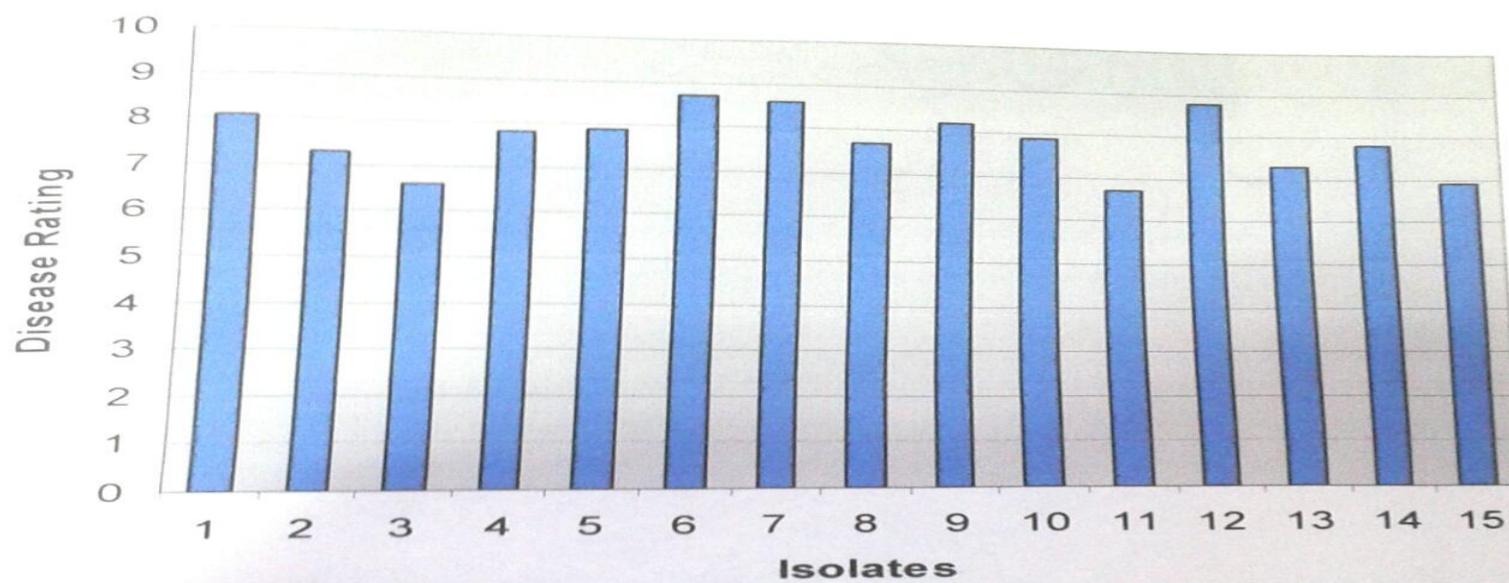
Observation on disease rating was recorded after 8th day of incubation on 20 seedlings. The isolates varied in causing the infection on variety JG 365, as it is evident from the data presented in table 14. On the basis of disease score the isolates were categorized into two groups viz., virulent and highly virulent. The disease rating of isolates were varied from 6.6-8.8. Out of 15 isolates 6 (40%) isolates were found virulent, as the disease rating score was 6.6-7.7 whereas 9 (60%) isolates categorized as highly virulent where the disease rating was observed ranging from 7.8-8.8. Two pathotypes were identified on the basis of pathogenicity test. Among all the isolates, the isolate no. belong to Dimni had maximum Disease

rating 8.8 while the minimum Disease rating 6.6 was observed in isolate no. 3 which is belong to Utila (Table 14) (Plate-10, Fig-2).

Table 15: Grouping of isolates of *Rhizoctonia bataticola* on the basis of pathogenicity

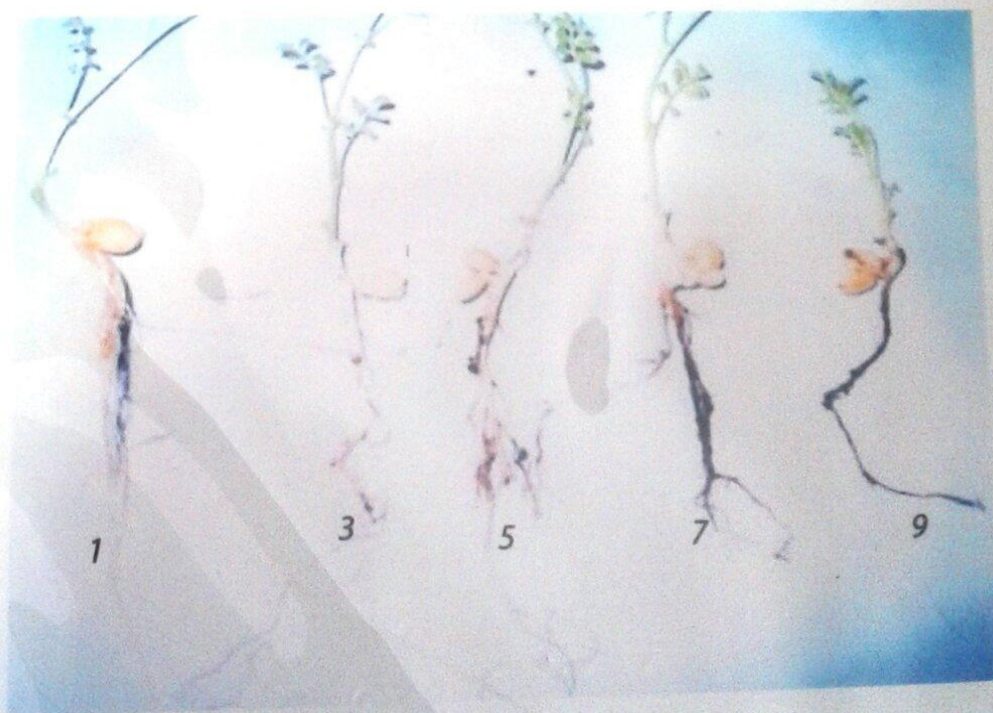
Category	Isolates	Total	Per cent
Virulent	2, 3, 8,11,13,15	6	40
Highly virulent	1, 4, 5,6,7,9,10,12,14	9	60
Total		15	100

Fig. 2: Reaction of Different Isolates of *Rhizoctonia bataticola* on chickpea variety JG-365





Pathogenicity test for *Rhizoctonia bataticola* .



Disease Rating 1-9

CHAPTER – V

DISCUSSION

In Indian agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Chickpea is an important pulse crop of the Indian subcontinent. The crop suffers due to many diseases of which dry root rot caused by *Rhizoctonia bataticola* is the most important as heavy losses are caused by this disease. The diseases cause plant mortality in podding and maturity stage of the crop growth when the temperature exceeds >25-30°C.

Present investigations were undertaken with a view to know the variability among isolates of *R. bataticola* (cultural, morphological, and pathological variability) collected from survey of 15 locations of Gwalior, Bhind and Morena districts of Madhya Pradesh. A very little information on variability among isolates of *R. bataticola* is available in the literature particularly in chickpea dry root rot.

During collection of the dry root rot infected plants, the disease was found to occur at flowering and podding stage of the crop in the range of 5.66 to 22.77 per cent. The dry root rot symptoms observed in diseased plant exhibited straw color appearance and taproot mainly becomes brittle having minute sclerotia. These symptoms separated this disease from the so-called “Wilt complex “as stated by various workers in India. (Nene, 1989). Maximum atmospheric temperature up to 34.5°C favored the disease development. Substantial mortality and loss in chickpea due to *R. bataticola* had been reported in high ambient temperature 30°C and above in the post – flowering stage (Singh *et al* 1990).

Fifteen isolates as pure culture of *R. bataticola* were isolated from infected plants / root sample from Gwalior, Bhind and Morena districts of Madhya Pradesh and were designated, as isolates number 1 to 15. Isolates of *R. bataticola* were grouped on the basis of cultural, morphological and pathological characters.

Cultural characters of all the isolates of *R. bataticola* on P.D.A. medium showed variation within the isolates. Grouping of isolates on the basis of type of growth *viz.*, fluffy, submerged and partial submerged. Indicates the possibility of three groups within the isolates. In case of mycelium color, three types (brown, dark

brown, black) were found. Dark brown type of mycelium being the highest as the maximum number of isolates (46.67%) belongs to this category. These results confirm the observations made by Byadgi and Hegde (1985) who reported fluffy, dark brown mycelium, partial fluffy mycelium and appressed light brown mycelium growth on different isolates of *R. bataticola*. Other workers have also reported the variation in type of growth isolates of *R. bataticola* (Sobti and Bansal, 1988; Ratnoo *et al.* 1997; Hooda and Grover, 1982)

After critically going through the data on growth rate recorded after 24^{hr} intervals concluded that the isolates were fast growing (5 isolates), medium growing (6 isolates) and slow growing (4 isolates), which indicate differential growth rate in isolates. The observations coincide with Jharia (1985) who observed variation in growth rate of *R. bataticola*. Monga and Sheo Raj (1994) who observed that Rs-2 group of isolates was fastest in mycelial growth and Rs-4 was slow growing among all the four groups of isolates on potato dextrose agar medium. Byadgi and Hegde (1985) also reported fast growth and slow growth in his investigation.

Morphological characters of all the isolates showed variations within isolates. The size of hyphal width grouped as very thin, thin and thick which ranged from 3.0-5.7µm. thin type of hyphal width being the highest as the maximum number of isolates (46.67%) belongs to this category. Present observations coincide with the observations of Sobti and Sharma (1992) who reported that the width of hyphae varied from 4.16-8.48µm. Jharia (1985) observed the hyphal width range between 3.00-4.50µm.

The size of sclerotia of the isolates also has a wide range of variation from 27.0-90.0 x 24.0-81.6µm and thus categorized as minute, medium and bold type. Minute type being the highest, as the maximum number of isolates 46.66% belongs to this category. The above findings were also reported that the sclerotial size range between 66.14-128.25µm Mandal *et al.* (1998), 122-188 x 77-127µm from root isolates. Dhingra and Sinclair (1973), 84-93 µm Jharia (1985), 73.88-188.33 x 51.09-155.90 µm Sobti and Sharma (1992), 60-165 X 57-114 µm Critopoulos (1953), 61 - 90 µm (Raut and Ingle, 1989).

On the basis of number of sclerotia per microscopic field the isolates were grouped into three groups as minimum (10 isolates), medium (4 isolates) and

maximum (1 isolates) number of sclerotia. The number of sclerotia ranged from 6.7 to 55. In the present finding 66.67% of the isolates were having minimum number of sclerotia these results are confirmed with the findings of Shekhar *et al.* (2006) who reported that Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia / 9 mm disc and 52 / microscopic 10X field) where as Coimbatore isolate produced minimum number of sclerotia (169 sclerotia).

Color of sclerotia of all the isolates were varied and thus categorized as brown (3 isolates), dark brown (7 isolates) and black (5 isolates). The above findings were also reported by Mandal *et al.* (1998). He reported light to dark brown sclerotia. Mandal *et al.* (1998) and Ratnoo *et al.* (1997) reported light to dark brown sclerotia in his investigation while Byadgi and Hegde (1985) reported brown to black color sclerotia in his investigation.

Shape of sclerotia was also observed for each isolates and categorized into two groups as oval shape and round shape. Out of the 15 isolates 4 (26.67%) Isolates were found as oval shape and 11 (73.33%) isolates were found as round shape of sclerotia. The above findings was also reported by Shekhar *et al.* (2006) they found that oblong shape having irregular edges and round shape with regular edges of sclerotia. Mandal *et al.* (1998) reported irregular and round to elongate sclerotial shape.

Pathological variability has been established by inoculating 5 days old seedling of JG-365 from all the 15 isolates of *R. bataticola* by using blotter paper technique. On the basis of variation exists in their virulence leading to the development of disease rating (1-9 scale), the isolates may be grouped into 2 categories viz., virulent (6 isolates), and 9 isolates as highly virulent. Highly virulent being the highest, as the maximum number of the isolates (60%) belongs to this category. Shekhar *et al.* (2006) reported variation in pathogenic behavior of *M. phaseolina* and grouped as most virulent and least virulent isolates. Byadgi and Hegde (1985) studied and classified 7 isolates into three groups; A (Virulent) B (Intermediate) and C (Mild) on the basis of pathogenicity. Other workers have also reported the differential reaction by the isolates of *R. bataticola* in chickpea (Ratnoo *et al.* 1997; Singh *et al.* 1990; Monga and Sheo Raj, 1994) and found considerable variation among isolates.

Present studies conducted on various aspects of variability in the isolates of *Rhizoctonia bataticola* led to the conclusion that the isolates pertaining to dark brown mycelium (46.67), medium growth rate (40%), partial submerged (46.67%), thin hyphal width (46.67%), sclerotia minute size (46.66%), and dark brown (46.67%) and production of sclerotia (100%) is positively correlated with the virulence of as is reflected by the isolates number (1, 4, 5, 6, 7, 9, 10, 12, 14). The reaction produced by these isolates in susceptible variety (JG-365) is highly pathogenic giving disease rating of 8–9. Similar conclusion was drawn by Monga and Sheoraj (1994) in case of cottony root rot incited by *M. phaseolina*.

CHAPTER - 6

SUMMARY, CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

6.1 Summary

In Indian agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Chickpea is an important pulse crop of the Indian subcontinent. The crop suffers due to many diseases of which dry root rot caused by *Rhizoctonia bataticola* is the most important as heavy losses are caused by this disease. Present investigations were undertaken with a view to know the variability among isolates of *R. bataticola* (cultural, morphological, and pathological variability) collected from survey of 15 locations of Gwalior, Bhind and Morena districts of Madhya Pradesh.

Diagnosis of symptoms of disease at different stage of crop growth revealed the disease to be dry root rot. The symptoms were mainly brittleness of taproot and presence of minute sclerotia and on isolation yielded *R. bataticola*. (Taub.) Butler.

Variability among the isolates was further studied in respect to cultural morphological and pathological characterization. Cultural studies of all 15 isolates of *R. bataticola* revealed that isolates differ in the type of mycelial growth and growth rate on potato dextrose agar medium to be favorable for luxuriant growth for all the isolates.

Morphological studies of different isolates of *R. bataticola* revealed variability in size of hyphal width (3.0-5.7), size of sclerotia (27.0-90.0 x 24.0-81.6µm), number of sclerotia (6.7-55), on the basis of color of sclerotial measurement the sclerotia were categorized into three group *i.e.*, brown, dark brown and black; while looking to the shape of sclerotia, the isolates were categorized into two groups *i.e.*, round and oblong.

Pathogenicity test was conducted by inoculating JG-365 a dry root rot susceptible variety of chickpea with 15 isolates by using blotter paper technique. All the isolates exhibited clear-cut variation in their virulence and characterized as,

virulent and highly virulent. Indicating the possibility of two pathotypes and may be correlated them with the virulence of the isolates.

6.2 Conclusion

Chickpea (*Cicer arietinum* L.) is an important Rabi pulse crop in Madhya Pradesh. Studies on 15 isolates of *R. bataticola* obtained from chickpea in Gwalior, Bhind and Morena districts of Madhya Pradesh showed considerable variability in cultural, morphological and pathological characters and may be grouped into 2-3 pathotypes.

Present studies conducted on various aspects of variability in the isolates of *R. bataticola* led to be conclusion that the isolates having dark brown mycelium (46.67), medium growth rate (40%), partial submerged (46.67%), thin hypha (46.67%), sclerotia minute size (46.66%) and dark brown (46.67%) and production of sclerotia (100%) is positively correlated with the virulence.

6.3 Suggestions for further work

- Pathogenic variation in different isolates of *R. bataticola* by using differentials for known genotypes / varieties needs to be further investigated under lab and field conditions.
- Variations in different isolates of *R. bataticola* on their molecular basis need to be worked out.

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APPENDIX-I

Analysis of variance (ANOVA)

Growth rate of different isolates of *Rhizoctonia bataticola* at different intervals

Source of Variations	d.f.	Sum of squares	Mean sum of squares	F Ratio
Treatments	14	10377.2	741.22	1.40
Error (A)	60	34367.6	527.79*	
Total	74			

VITA

The author of the thesis Vishal Gurav S/o Shri Shivaji Gurav was born on 1st June, 1987 at Distt. Satara (M.S.).

I passed Matriculation (X) in the year 2002 from Maharashtra state Board, Pune with 72.40 per cent marks. Passed Intermediate (XII) in the year 2004 from Maharashtra state Board, Pune with 62.83 per cent marks.

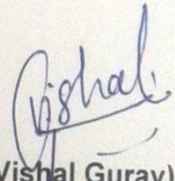
I joined Krishna College of Agriculture, Rethare Bk. in the year 2004 and was awarded B.Sc. (Ag.) degree in the year 2008, achieving OGPA 6.96 out of 10 scale from MPKV, Rahuri.

After completing graduation I was selected for M.Sc. (Ag.) degree programme in Plant Pathology through ICAR and, was admitted in RVSKVV, Gwalior, College of Agriculture, Gwalior (M.P.). I completed all required courses of M.Sc. (Ag.) by achieving approximately OGPA 7.90 out of 10 scale and worked for my M.Sc. (Ag.) thesis under the guidance of Dr. (Mrs.) Reeti Singh.

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Date : 16-11-10

Place : Gwalior


(Vishal Gurav)